

# ICM 42688 IMU 0 | SPI 1

- User Bank 0 Registers

| Addr (Hex) | Addr (Dec) | Register Name     | Data               |
|------------|------------|-------------------|--------------------|
| 14         | 20         | INT_CONFIG        | 0b110              |
| 1D         | 29         | TEMP_DATA1        | TEMP_DATA[15:8]    |
| 1E         | 30         | TEMP_DATA0        | TEMP_DATA[7:0]     |
| 1F         | 31         | ACCEL_DATA_X1     | ACCEL_DATA_X[15:8] |
| 20         | 32         | ACCEL_DATA_X0     | ACCEL_DATA_X[7:0]  |
| 21         | 33         | ACCEL_DATA_Y1     | ACCEL_DATA_Y[15:8] |
| 22         | 34         | ACCEL_DATA_Y0     | ACCEL_DATA_Y[7:0]  |
| 23         | 35         | ACCEL_DATA_Z1     | ACCEL_DATA_Z[15:8] |
| 24         | 36         | ACCEL_DATA_Z0     | ACCEL_DATA_Z[7:0]  |
| 25         | 37         | GYRO_DATA_X1      | GYRO_DATA_X[15:8]  |
| 26         | 38         | GYRO_DATA_X0      | GYRO_DATA_X[7:0]   |
| 27         | 39         | GYRO_DATA_Y1      | GYRO_DATA_Y[15:8]  |
| 28         | 40         | GYRO_DATA_Y0      | GYRO_DATA_Y[7:0]   |
| 29         | 41         | GYRO_DATA_Z1      | GYRO_DATA_Z[15:8]  |
| 2A         | 42         | GYRO_DATA_Z0      | GYRO_DATA_Z[7:0]   |
| 2D         | 45         | INT_STATUS        |                    |
| 4B         | 75         | SIGNAL_PATH_RESET | 0b100              |
| 4E         | 78         | PWR_MGMT0         | 0b010011           |
| 4F         | 79         | GYRO_CONFIG0      |                    |
| 50         | 80         | ACCEL_CONFIG0     |                    |
| 54         | 84         | TMST_CONFIG       | 0b10011            |
| 63         | 99         | INT_CONFIG0       | 0b00100000         |
| 65         | 101        | INT_SOURCE0       | 0b00011000         |
| 70         | 112        | SELF_TEST_CONFIG  |                    |
| 75         | 117        | WHO_AM_I          |                    |
| 76         | 118        | REG_BANK_SEL      |                    |

## • User Bank 1 Registers

| Addr (Hex) | Addr (Dec) | Register Name | Data                  |
|------------|------------|---------------|-----------------------|
| 5F         | 95         | XG_ST_DATA    | X-gyro self test data |
| 60         | 96         | YG_ST_DATA    | Y-gyro self test data |
| 61         | 97         | ZG_ST_DATA    | Z-gyro self test data |
| 62         | 98         | TMSTVAL0      | TMST_VALUE[7:0]       |
| 63         | 99         | TMSTVAL1      | TMST_VALUE[15:8]      |
| 64         | 100        | TMSTVAL2      | TMST_VALUE[19:16]     |

## • User Bank 2 Registers

| Addr (Hex) | Addr (Dec) | Register Name | Data                   |
|------------|------------|---------------|------------------------|
| 3B         | 59         | XA_ST_DATA    | X-accel self test data |
| 3C         | 60         | YA_ST_DATA    | Y-accel self test data |
| 3D         | 61         | ZA_ST_DATA    | Z-accel self test data |

## • Characteristics

### ◦ Gyroscope

- Gyroscope range: 2000 dps
- Sensitivity scale factor for 2k dps: 16.4 LSB/dps
- Initial zero rate output tolerance: +- 0.5 dps
- Gyroscope start up time: 30 ms
- ODR: 1 kHz

### ◦ Accelerometer

- Accelerometer range: 16 g
- Sensitivity scale factor for 16 g: 2048 LSB/g
- Initial zero rate output tolerance: +- 20 mg
- Accelerometer start up time: 10 ms
- ODR: 1 kHz

- Temperature in Degrees Centigrade =  $(\text{TEMP\_DATA} / 132.48) + 25$

## • SPI

- Pull CS to activate: LOW
- Data delivered MSB first LSB last
- Data transitioned on falling edge of SCLK

- Self test at boot for later version
- Interrupts
  - INT fire pull: LOW
  - Use INT 1
  - INT fired when DATA READY, cleared when DATA READ
- Time stamp of reading stored in register, resolution of **1 us**
- Power mode: 6-axis low-noise mode